

Research Programs in Catalysis: Prior Work, Original Questions, and Experimental Decisions

September 2026

Abstract

This document proposes four experimental research programs in heterogeneous catalysis, ranked for development, and briefly records six reserve ideas. It reports no new experiments. Its results are a check of each idea against prior work, calculations from published data that size what is at stake, and experimental designs whose rules for expanding or stopping are fixed in advance.

(1) *Physical water management.* Promotion by physically mixed hydrophobic polymers is established. On a Co/SiO₂ Fischer–Tropsch catalyst, a small polydivinylbenzene (PDVB) addition kept CO conversion near 44% for 1200 h while a quartz-diluted reference declined. From the published data we calculate a C₅₊ output proxy over 25–585 h of 1.89 times the reference (1.80 per unit catalyst-plus-polymer mass). We also show with a simple storage model that faster outlet-water clearance alone cannot show lower water exposure at active sites. New here: does PDVB added to a catalyst that has already run without it protect the catalyst against a later water challenge, and, conditionally, do measured responses predict output during a small reversible humidity pulse?

(2) *Cyclic oxides.* Li₂CO₃-coated La_{0.8}Sr_{0.2}FeO₃ is a selective oxygen carrier for chemical-looping oxidative dehydrogenation of ethane and was cycled for 1000 h with Ar purges. The accompanying process model replaces those purges with steam, which was not tested on the coating; CO₂ regeneration of carbonate melts has close precedents. New here: does steam in the post-oxidation purge change ethylene output, and is CO₂ supplied with the steam (protection) better than CO₂ supplied afterwards (recovery)? As an illustration, fully recovering the published Ar-aged C₂₊ yield loss would pay for at most 2.02 extra minutes per 16-minute cycle.

(3) *Polymer ethenolysis.* One fresh Na/alumina addition restores the output of a reused Na/alumina–W/silica mixture that converts polyethylene to propylene; pressure effects and chain-population models are also prior work. New here: can a lower ethylene partial pressure let a smaller fresh Na/alumina addition give the same output, and can that saving be predicted? At equal output, halving it would use 25% less fresh Na/alumina and 16.7% less total fresh catalyst solid.

(4) *Promoted Ag.* Better selectivity retention by Ni-containing Cs/Re-promoted Ag was disclosed in 1995, confounded by Cs and treatment differences, and chloride adjustment alone changes selectivity and output. New here: a composition-controlled test of when Ni adds ethylene oxide output that chloride operation cannot replace, and a prediction of when a chloride recovery step pays. At fixed output, raising selectivity from 0.90 to 0.91 would cut reaction-level oxygen use by 4.40% and CO₂ formation by 10.99%.

The rank reflects scientific value: water management and cyclic oxides are the first and second priorities, although polymer ethenolysis and promoted Ag may be quicker first experiments where equipment exists. A program expands only if its measurements predict useful output under a condition withheld from fitting.

Contents

Overview and recommended priorities	3
1 Physical water management: when does a neighboring hydrophobic material protect a catalyst?	6
1.1 What is known and what remains open	6
1.2 Why outlet water transients cannot locate active-site water exposure	7
1.3 Experiment 1: qualify the platform and the late-addition procedure	7
1.4 Experiment 2: distinguish protection from an activity penalty	8
1.5 Experiment 3: predict output under a withheld humidity pulse	9
1.6 Decisions and stop rules	10
2 Cyclic oxides: steam compatibility and recovery of selective oxygen delivery	11
2.1 Prior work and the open question	11
2.2 Chemical hypothesis and competing explanations	12
2.3 Experiment 1: ordinary steam versus Ar cycling, then a timing diagnostic	12
2.4 Common reset, function assay, and inventories	13
2.5 Experiment 2: a usable management policy and its complete-cycle value	15
2.6 Experiment 3: predicting recoverability under a new operating history	15
2.7 Outcomes and decisions	16
3 Polymer ethenolysis: less fresh catalyst during reuse	17
3.1 Prior work	17
3.2 Carbon origin and catalyst demand	18
3.3 Experiment 1: reproduce reuse and bracket the fresh Na/alumina dose	18
3.4 Experiment 2: one lower ethylene pressure crossed with dose	19
3.5 Chain-state evidence and selective-initiation claims	20
3.6 Experiment 3: predict a new dose, then test whether the saving persists	21
4 Promoted Ag: when does dilute nickel add useful ethylene oxide output?	22
4.1 Prior work and the remaining question	22
4.2 Stage A: validate the materials and define a fair output comparison	23
4.3 Stage B: compare ordinary drift and operating policies	24
4.4 Stage C: test whether the difference exceeds the known chloride response	25
4.5 Stage D: predict when a chloride recovery step pays off	26
5 Reserve ideas and withdrawn claims	28
References	30

Overview and recommended priorities

Purpose and scope. This document recommends four research programs for development, in the order of Table 1. Each links an important catalytic limitation to a specific intervention, a fair material or operating comparator, and a prediction that independent experiments can test. Four is an editorial choice: it gives full treatment to the two strongest development candidates and to two narrower studies that each end in a concrete decision about resource use. Six reserve ideas, whose current limitations do not justify the same experimental scope, are summarized in Section 5.

Table 1: The four programs in rank order: what prior work already shows, the open question each addresses, and the reason for its rank.

Program	Already known	Open question (contribution sought)	Reason for rank
1. Physical water management (Section 1)	Physically mixed hydrophobic polymer promotes CO and CO ₂ hydrogenation.[1, 2] With a small PDVB addition, Co/SiO ₂ kept about 44% CO conversion in Fischer–Tropsch synthesis for 1200 h; a quartz-diluted reference started higher but declined strongly.[3]	Does PDVB added after common conditioning without it protect against a later water challenge at one catalyst state? Conditionally, do measured responses predict useful output during a small reversible humidity pulse?	A large published formulation effect. The pulse prediction would not establish a damage law across histories.
2. Cyclic oxides (Section 2)	Li ₂ CO ₃ -coated La _{0.8} Sr _{0.2} FeO ₃ (LSF) gives selective chemical-looping oxidative dehydrogenation of ethane and was cycled for 1000 h with Ar purges.[4, 5] Carbonate/hydroxide response to steam and CO ₂ regeneration have close precedents.[6–8]	Does steam in the post-oxidation purge change ethylene output, and is CO ₂ supplied with the steam (protection) better than CO ₂ supplied afterwards (recovery)?	A specific gap between an Ar-purged durability test and a steam-based process proposal.
3. Polymer ethenolysis (Section 3)	Na/alumina with W/silica converts polyethylene to propylene; one fresh Na/alumina addition restores output during reuse.[9, Fig. 4B] Pressure effects and chain-population models are prior work.[10–12]	Can one lower ethylene partial pressure replace part of the fresh Na/alumina while keeping polymer-derived output, and can the saving be predicted?	A concrete catalyst-use decision, although rescue and pressure effects have close prior art.
4. Promoted Ag (Section 4)	Ni-containing promoted Ag retained selectivity better than a Ni-free parent in a 1995 patent, with Cs and treatment differences between them.[13] Dilute Ni suppresses combustion more than ethylene oxide formation on Ag lacking the full Cs/Re promoter system, and chloride operation strongly changes selectivity and rate.[14–16]	When does Ni add useful ethylene oxide output that chloride adjustment cannot replace, and can that be predicted?	Historical Ni-retention evidence leaves a demanding, narrow material and operating study.

Main results of this analysis. The following calculations and corrections are this document’s own results; they use published data and are not new measurements.

- *Water management.* Integrating Fang et al.’s published conversion and selectivity series over 25–585 h gives a C₅₊ output proxy 1.89 times the reference, or 1.80 per unit catalyst-plus-polymer mass (Eq. 1). This is not a carbon balance or a lifetime extrapolation. A two-pool storage model shows that outlet water transients cannot locate where water is produced or held, so faster clearance alone does not show drier active sites.
- *Cyclic oxides.* As an illustration, fully restoring the published Ar-aged C₂₊ yield (47.24% to 53.2%) would pay for at most 2.02 additional minutes per 16-minute cycle, before gas and heat costs (Eq. 12). This bounds the value of any recovery step; it does not predict steam damage.
- *Polymer ethenolysis.* At equal output, a one-time 0.2 g makeup would use 25% less fresh Na/alumina and 16.7% less total fresh catalyst solid than the published 0.4 g addition (Table 5).
- *Promoted Ag.* If ethylene oxide and complete combustion are the only carbon outlets, raising selectivity from 0.90 to 0.91 at fixed ethylene oxide output reduces reaction-level ethylene use by 1.10%, oxygen use by 4.40% and CO₂ formation by 10.99%.
- *Withdrawn claims.* A faster outlet-water transient does not by itself show lower active-site water exposure; fresh-Na rescue does not identify selective loss of initiation; Ni compatibility and retention on promoted Ag are prior art; and published mannose results contradict treating mannose as an inactive reservoir in tungsten sugar conversion (Section 5).

Priorities and feasibility. The ranking concerns scientific development and space in the document, not a plan to run four campaigns at once. Water management is the first priority and cyclic oxides the second. Polymer ethenolysis and promoted Ag may nevertheless offer the best first experiment where their reactors, reference materials and analytical methods already exist. Laboratory access, preparation reproducibility, achievable uncertainty and actual costs have not been established. None of the four yet justifies an open-ended campaign.

Three kinds of statement. The document keeps three kinds of statement separate. *Published observations* are attributed to their sources and keep the sources’ preparation, measurement and time limits. *Calculations* are deductions from stated assumptions or identified published data; they are not new measurements. *Proposed experiments* have not been performed; each states the information a decision needs, the possible outcomes and the limits of interpretation. The contribution is a testable research design, not advance confirmation of a mechanism. In particular, the document does not establish lower local water exposure at Co sites, steam-induced Li loss, selective loss of polymer initiation, or preservation of a particular Ag–Re oxygen-transfer state.

When to expand or stop a program. Each design compares against a fair comparator and a simpler credible operating alternative: PDVB versus quartz, both added after common conditioning; ordinary steam versus Ar before any timed CO₂ treatment; ethylene pressure crossed with fresh-Na dose; and Ni-treated versus sham-processed Ag under ordinary and recovery policies, plus untreated Ag under ordinary operation. A program expands only when its measurements identify a material or operating choice and correctly predict that choice’s response under a condition or history withheld from fitting. Further characterization is justified only when it separates explanations that would lead to different decisions. Output accounting stays complete even when a mechanistic interpretation succeeds.

Design rules common to all programs. The chapters apply the following rules without restating them each time.

- *Replication.* Replicates are independent catalyst charges or specimens, randomized within preparation and run-order blocks. Repeated analyses or cycles on one bed are not replicates. Replicate and cycle counts come from pilot variability and the smallest important difference, both fixed before the decisive runs.
- *Null results.* A null result counts as evidence against an important effect only if its uncertainty interval (including preparation, run, calibration and balance uncertainty) excludes the smallest important difference. Positive and null results alike hold only for the tested material, conditions, duration and history; no finite laboratory campaign here can establish lifetime tolerance or competitive industrial lifetime.
- *Predictions.* Models, predictions and their uncertainty intervals are fixed before the withheld test, which uses independent charges. A failed prediction is reported as a failure, not refitted. Where a chapter names a simpler baseline, a more detailed model or added mechanism is credited only if it predicts the withheld result better than that baseline; otherwise only the simpler rule is claimed.
- *Accounting.* Practical output comparisons use complete elapsed time over a horizon fixed in advance, counting every treatment, handling and recovery step that differs between arms, with all catalyst, gas and product inventories closed. Absolute outputs are reported alongside ratios or fractions.
- *Ex situ states.* Characterization after quench or transfer is interpreted only after showing that sampling does not change the measured property; otherwise the attribution is withheld.

1 Physical water management: when does a neighboring hydrophobic material protect a catalyst?

Summary. Cobalt Fischer–Tropsch synthesis converts CO and hydrogen (syngas) to hydrocarbons and water; C_{5+} denotes hydrocarbons with at least five carbon atoms. On graphite-supported cobalt, externally supplied water at very low conversion, followed by a dry return, has been linked to oxidation, sintering, and lost function.[17] Physically mixing hydrophobic polydivinylbenzene (PDVB) with a catalyst has promoted CO hydrogenation on Co–Mn carbide and CO₂ hydrogenation on copper.[1, 2] On Co/SiO₂, Fang and colleagues (2026) found that 0.025 g PDVB per 0.5 g catalyst kept CO conversion near 44% and C_{5+} selectivity at 88–90% for 1200 h, while a quartz-diluted reference declined strongly.[3] From their Source Data, we calculate a C_{5+} output proxy over 25–585 h of 1.89 times the reference’s (1.80 per unit catalyst-plus-polymer mass). A simple storage model shows why outlet water transients alone cannot identify active-site water exposure. What is new is adding PDVB late, after the catalyst has run without it (common conditioning), and asking whether it still protects the catalyst and whether measured water and catalytic responses predict output under a new disturbance. We propose comparing late PDVB with matched quartz under a controlled water challenge with time-matched controls and, conditionally, predicting C_{5+} output under a withheld humidity pulse. The outcome would be a controlled protection test at one catalyst state and disturbance and, if prediction succeeds, a predictive rule within the calibrated range.

1.1 What is known and what remains open

PDVB promotion of Co–Mn carbide in syngas conversion showed mixing-distance and water-cofeed effects, and the carbide recovered after PDVB removal performed comparably to the original; physical promotion also works for copper-catalyzed methanol synthesis.[1, 2] Physical mixing, post-use additive removal, transfer to another reaction, and contact-angle correlations are therefore not new.

Fang’s Co/SiO₂ test ran at 220 °C and 2 MPa with separately prepared 40–60 mesh granules; the quartz-diluted reference, at matched nominal bed volume, initially converted more CO. Water-cofeed, heat-transfer, material-state, and mixing controls support a substantial functional effect,[3] but a mechanism requires separating retained water, water transport, reaction rate, and catalyst history. The output proxy integrates CO conversion X_{CO} times the published carbon-based C_{5+} selectivity $S_{C_{5+}}$ over the common 25–585 h interval,

$$Q^* = \int X_{CO}(t)S_{C_{5+}}(t) dt = 116.0 \text{ h (reference), } 219.8 \text{ h (PDVB),} \quad (1)$$

using sheet “Figure 1” of Fang’s Source Data workbook (equal nominal CO feed), piecewise-linear interpolation of each series, and the combined time grid. The 1.80 ratio divides by the 5% larger catalyst-plus-polymer mass; quartz is not counted. The proxy omits the first 25 h and is not a complete carbon balance, an uncertainty interval, or a lifetime extrapolation; it does not show superiority to an optimized cobalt formulation or regeneration policy.[3]

Water history and pore environment also have precedents: water pretreatment versus later cofeed and dry return on alumina-supported cobalt;[18] activation-water effects on later cobalt performance;[19] silica-shell geometry and deactivation, separating water-related from heavy-hydrocarbon-related behavior;[20] and strongly retained oxygenates that change water affinity during operation,[21] so a dry fresh-material contact angle cannot be assumed to represent a working catalyst. What is new here is *late addition after common conditioning*.

1.2 Why outlet water transients cannot locate active-site water exposure

Faster water clearance alone does not show that active sites see less water. Surfaces of different wettability hold different amounts of water at one chemical potential, but this alone creates no steady gradient across an otherwise equilibrated, connected gas; only reaction, imposed flow, or another source or sink can. Wettability then acts through storage, liquid configuration, and transport resistance, so equilibrium loading differences alone cannot explain continuous drying.

One pool: a shorter transient need not mean lower exposure. A shorter transient can come from less storage or from faster exchange; only faster exchange lowers the steady water excess:

$$B \frac{dc}{dt} = q - G(c - c_b), \quad \tau = \frac{B}{G}, \quad c_{ss} - c_b = \frac{q}{G}. \quad (2)$$

Here c and c_b are local and boundary concentrations expressed as equivalent gas-phase concentrations, B is differential storage capacity (volume), G is exchange conductance (volume/time), and q is water production (amount/time), for a locally linear, time-invariant compartment with a specified source. Smaller B leaves the steady excess unchanged and can raise exposure early in an upward step while shortening the tail, so less storage need not improve durability. Fang’s syngas-to-Ar decay times, about 80 s for the PDVB mixture and 175 s for Co/SiO₂ alone, cannot separate these cases, because that switch also changes water production and adsorbed species.[3, Fig. 3d]

Two pools: the outlet does not locate the source. Even absolute inventory plus several outlet transients need not identify where water is produced. Two pools have concentrations c_i , capacities B_i , and conductances $G_i = \lambda B_i$ (λ : common exchange rate constant, inverse time); $B_T = B_1 + B_2$ and $W = B_1 c_1 + B_2 c_2$. With total water production q_T , inlet concentration c_{in} , a well-mixed gas volume V at concentration c_g , and actual volumetric flow F ,

$$\dot{W} = \lambda(B_T c_g - W) + q_T, \quad (3)$$

$$V \dot{c}_g = F(c_{in} - c_g) + \lambda(W - B_T c_g). \quad (4)$$

Of the pool properties, only B_T , q_T , and the aggregate inventory W enter the outlet c_g . If pool 1 produces all the water, its steady excess over the gas, $c_{1,ss} - c_{g,ss} = q_T/(\lambda B_1)$, falls threefold when B_1 moves from $B_T/4$ to $3B_T/4$, with no change in any outlet response for identical aggregate initial conditions. (The pool-to-inlet excess does not generally scale threefold, because it includes the unchanged $c_{g,ss} - c_{in} = q_T/F$.) This constructed example, not fitted to PDVB, shows that a correct outlet fit establishes only a formulation response. An active-site claim needs a source-specific or spatial constraint, which isotope signals do not automatically supply. A stationary transport explanation must also locate the relevant resistance: changing an external resistance cannot remove an unchanged internal one, and granule size alone cannot separate them.

1.3 Experiment 1: qualify the platform and the late-addition procedure

Reproduce the low-dose, separate-granule contrast with measured inventory, bed volume, temperature, pressure drop, carbon balance, and useful-product rate; the source’s powder co-granulate imaging and some high-dose controls do not describe this packing. The source restart is incompletely specified (synthesis includes reduction and passivation, but Supplementary Note 10 calls the standard test hydrogen-pre-reduced), so establish one reproducible activation sequence rather than silently adopting the 300 °C analytical pretreatment.[3]

Condition independently preweighed charges from one preparation, each keeping its identical initial quartz dilution throughout handling, without PDVB in syngas (initially $\text{H}_2/\text{CO}/\text{Ar} = 64/32/4$, 220°C , 2 MPa). A pilot on the parent (PDVB-free) Co/SiO_2 catalyst sets the duration beyond startup from rate and selectivity drift over a defined observation window, with tolerances from pilot variability; fix it and the complete operating schedule before randomization, and report pre-handling drift and retained liquid. This defines an operationally conditioned age, not microscopic stationarity or stable water affinity.

Assign charges randomly to the four conditions in Table 2. Recover each complete catalyst-and-quartz zone under inert gas with its retained liquid and, after showing quantitative solid recovery, normalize to the original dry catalyst inventory and record liquid loss separately. Cooling, transfer, and restart are identical for all charges, with no solvent wash or new reduction before the challenge.

Table 2: The main late-addition comparison; each condition uses independent catalyst charges.

After common conditioning	Water challenge	Time-matched control (sham)
0.05 g PDVB/g catalyst	Low–high–low water	Low water throughout
Matched quartz diluent	Low–high–low water	Low water throughout

Repack each whole charge with 0.05 g PDVB/g catalyst or the same measured bulk volume of quartz, to equal final bed volume, with fixed granule size, no selective removal of initial diluent, and measured void volume and pressure drop in every repacked bed rather than assumed equal hydrodynamics. A handling qualification pilot compares restarted parent charges with an uninterrupted parent control (packing, heat and flow behavior, solid and liquid recovery). A reproducible shift can define the post-handling starting state; uncontrolled solid or liquid loss or an erratic restart ends the late-addition work. Handling cannot recreate uninterrupted operation; this limits interpretation.

1.4 Experiment 2: distinguish protection from an activity penalty

Select one water challenge in a parent-only feasibility study, starting from 220°C and 20 bar (2 MPa) total with inlet $p_{\text{H}_2} = 10$ bar and $p_{\text{CO}} = 5$ bar, and water raised from 0.5 to 4 bar by replacing Ar at fixed inlet molar flow. These are design choices, not a demonstrated damage threshold for Co/SiO_2 . Fix the plateau and recovery durations before the four-condition comparison; the plateau must outlast measured equilibration and leave measurable but noncatastrophic residual loss in the parent. If no such disturbance exists within the qualified operating range, stop the damage test.

Target at most 2% CO conversion in the more active arm. With one water per CO converted,

$$\frac{\dot{n}_{\text{water,gen}}}{\dot{n}_{\text{water,in}}} \simeq \frac{X_{\text{CO}} y_{\text{CO,in}}}{y_{\text{water,in}}} = 0.025 \text{ at 4 bar water,} \quad 0.20 \text{ at 0.5 bar water,} \quad (5)$$

so external water dominates far more on the high plateau than during recovery; balances (CO_2 , oxygenates, gas contraction, unresolved oxygen storage) give the actual ratios. Low conversion limits whole-bed reactant changes, not necessarily intragranular gradients; before fixing the flow, check that a modest flow increase leaves the mass-normalized diagnostic rate materially unchanged. For equipment sizing only, and assuming the source’s nominal promoted CO consumption persists (not a rate prediction), roughly $68 \text{ L (g catalyst h)}^{-1}$ is needed for 2% conversion at inlet CO fraction 0.25: for a 50 mg charge, about 3.4 standard L/h of gas and 0.54 g/h of supplied water at 20% steam (0°C volume convention). Reducing inventory makes the test practical without reducing production per granule.

For each charge, measure C_{5+} carbon-production rates before the challenge and after recovery (r_{pre} , r_{post} , after defined settling windows). With $R = r_{\text{post}}/r_{\text{pre}}$, the primary protection contrast is

$$D = (R_{\text{PDVB,challenge}} - R_{\text{PDVB,sham}}) - (R_{\text{quartz,challenge}} - R_{\text{quartz,sham}}). \quad (6)$$

A positive D means that PDVB reduces the fractional loss of function associated with the challenge, beyond time-matched drift. The usefulness endpoint is the difference in integrated C_{5+} carbon output per complete bed volume over the fixed restart–challenge–recovery horizon, including the initial activity penalty, startup product inventories, and collection delays; report catalyst-plus-additive mass productivity separately. Because PDVB stays in the bed during recovery, better retention shows that the complete formulation works, not how many Co sites survive or that the avoided loss would have been irreversible.

A positive D after late addition rules out an explanation confined to the original conditioning, not effects during restart, the water transient, or ongoing liquid or chemical changes. The alternative that PDVB acts only by changing how much reaction water the bed makes is rejected only if parent-only runs span both formulations’ measured water histories, with uncertainty, in both challenge and recovery; the plateau alone is not enough, because generated water matters relatively more during recovery. Otherwise keep that alternative and report the formulation contrast. A null result after successful reproduction and qualification limits the supported benefit of late addition in this state; it does not show that startup protection caused the published result.

1.5 Experiment 3: predict output under a withheld humidity pulse

Pursue prediction only if Experiment 2 or the reversible water response poses a resolvable functional question. Zero-length-column work constrains water exchange in silica gel from storage data and flow-dependent transients, but its passive, low-temperature coefficients do not transfer to a reacting, liquid-bearing cobalt bed.[22]

Qualify water-specific blanks (heated transfer lines, pressure reduction, detector memory) at operating temperature, pressure, and composition; an inert tracer gives gas delay but not water-specific storage. On conditioned parent and PDVB packings, apply small upward and downward humidity steps at two flows with unchanged inlet H_2 and CO partial pressures, plus a reduced-amplitude repeat and baseline returns, integrating absolute water rates and measuring product rates fast enough to bound the changing chemical water source. Correct lower uptake per total mass for dilution: for additive/catalyst mass ratio a , noninteracting uptake is $(u_{\text{cat}} + au_{\text{add}})/(1 + a)$ even at unchanged catalyst loading.

Fit the simplest response model consistent with both step directions, both flows, the source balance, and independent packings. A measured decay rate is not automatically G/B : in a reacting compartment, linearizing Eq. 2 with $q = q(c)$ gives $(G - \partial q/\partial c)/B$, for $B > 0$ and a stable stationary state with $G > \partial q/\partial c$ there. If source feedback, blank uncertainty, or a changing liquid state prevents a meaningful coefficient, drop the conductance claim and test only an empirical input–output model, whose coefficients are not local transport properties.

The withheld test is a water pulse shorter than full equilibration. Fix in advance a horizon from pulse onset (the available production window) and an acceptable C_{5+} carbon-output deficit relative to undisturbed operation, so the endpoint says whether that excursion stays within budget. Calibrate each formulation’s nearby reversible product response, including product-detector and liquid-collection delays, and choose a pulse duration for which surviving models predict importantly different deficits over that fixed horizon; distinct water traces are not enough, since different time

constants with equal steady gains can give the same deficit integrated through complete recovery. Predict the water trace and the deficit with an endpoint interval narrower than the smallest important difference. The baseline uses inlet humidity, the calibrated steady product response, and measured apparatus delays. Success supports an operating-history rule within the calibrated state and amplitude range; if transient characterization does not beat the baseline, claim only the empirical rule. Neither outcome validates a damage law at much higher water exposure.

1.6 Decisions and stop rules

The results decide whether a simple formulation change pays for its initial penalty under the tested conditions, and when faster water clearance is an inadequate selection criterion. Stop the affected line of work if the source effect cannot be reproduced, handling destroys interpretability, the challenge has no measurable dynamic range, or the predicted functional difference is below measurement resolution. If fractional protection is real but cumulative output is worse, report the tradeoff and claim no practical advance. A useful formulation result stands without an identified mechanism; a mechanism paper would need a constraint that locates the producing region and stays valid when the additive changes liquid and catalyst state. A well-fitted water curve without an important functional prediction does not justify a larger materials or isotope campaign. Commercial claims need a later complete-cycle comparison with an optimized catalyst, including regeneration, additive inventory, bed volume, and replacement time.

2 Cyclic oxides: steam compatibility and recovery of selective oxygen delivery

Summary. In chemical-looping ethane oxidative dehydrogenation, a cycled oxide acts as an oxygen carrier: it supplies oxygen to ethane and is then reoxidized. Gao and colleagues showed that a Li_2CO_3 coating makes $\text{La}_{0.8}\text{Sr}_{0.2}\text{FeO}_3$ (LSF) a selective carrier.[4] Brody and colleagues cycled coated LSF for more than 1200 elapsed hours; in the first 1000 h, run with Ar purges, average C_{2+} carbon yield (ethylene plus C_{3+}) fell from 53.2% over hours 0–300 to 47.24% over hours 600–1000. Their process model replaces the Ar purges with steam, which the study did not test on the coating.[5] Carbonate/hydroxide chemistry and CO_2 regeneration have close precedents (Barckholtz,[6] Chacko,[7, ch. 4.1] Fereres[8]); another demonstration that CO_2 stabilizes carbonate would not suffice, so the question is narrow: does steam in the post-oxidation purge change ethylene output from this nominal 10 wt% coating, can a loss be recovered, and does a remedy repay its time and gas? We propose first to compare ordinary cycling with steam-rich and Ar post-oxidation purges; if a difference warrants it, a diagnostic compares CO_2 given with steam against CO_2 given afterward, and a policy comparison counts complete-cycle output. Success would show whether the substitution is compatible, recoverable or worth managing, and whether a measured state predicts recovery. As an illustration only, fully restoring Brody’s Ar-aged C_{2+} yield would repay at most about 2.02 extra minutes per 16-minute cycle, before gas and heat are charged (Eq. 12).

2.1 Prior work and the open question

The ideal selective reduction of the carrier is



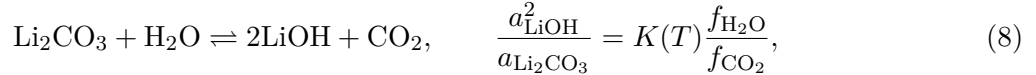
the challenge is to deliver enough oxygen for useful conversion without oxidizing hydrocarbons to CO or CO_2 . Gao and colleagues attributed the coated carrier’s selectivity to a molten layer that blocks nonselective sites and transports peroxide species associated with $\text{Fe}^{4+}/\text{Fe}^{3+}$ redox, and report about 0.42 wt% oxygen consumption over the first six minutes at 700 °C. The supported coating reportedly melts below bulk Li_2CO_3 , so hydroxide formation need not be what first makes it liquid.[4]

Brody and colleagues cycled 300 g of coated LSF. For the first 1000 h at 700 °C, each 16-minute cycle was 1 min ethane, 5 min Ar, 5 min oxidation and 5 min Ar. C_{2+} carbon yield averaged 53.2% over hours 0–300 and 47.24% over hours 600–1000; later temperature and flow changes improved performance, so the later optimum was not the condition used throughout the test. The methods give 10 wt% carbonate but the conclusion 20 wt%, so loading must be measured. The observed exothermic half-cycles do not show a continuously producing, thermally self-sufficient plant.[5]

Barckholtz and colleagues observed carbonate/hydroxide responses to steam and CO_2 in a mixed Li/Na melt, though their Raman signals gave no absolute concentrations.[6] Chacko cycled LSF with 45 wt% mixed carbonate at 800 °C and CO_2/O_2 regeneration, a close precedent for coupled coating/carrier regeneration despite the different formulation and temperature: carbonate decomposed during inert purges, and H_2 evolved during regeneration, attributed to reduced iron reacting with water released by hydroxide carbonation.[7, ch. 4.1] When Fereres and colleagues switched a bulk Li/Na/K carbonate melt from air to CO_2 after 30 days at 640 °C, further mass loss stopped but the initial state was not restored,[8] a precedent for prevention versus incomplete delayed recovery.

2.2 Chemical hypothesis and competing explanations

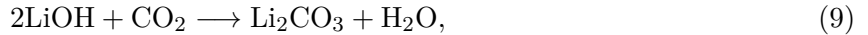
A candidate component equilibrium is



where a is condensed-phase activity and f fugacity (approximated by partial pressure in the atmospheric-pressure diagnostic). At fixed temperature a lower steam/ CO_2 ratio favors carbonate over hydroxide, but this direction gives no composition, equilibration time or Li-loss rate for supported LSF; nonideal melt chemistry, substrate interactions and changing coverage preclude importing a fuel-cell equilibrium constant uncritically. Barckholtz held each condition for hours, so those data do not show a response within a five-minute purge.[6]

Steam could convert the retained salt reversibly, redistribute it away from important interfaces, move Li onto reactor materials or into transported material, or change the carrier’s redox state or phase, singly or in sequence. Downstream Li does not identify LiOH vapor, unchanged total Li does not show unchanged coverage, and a smaller carbonate surface signal does not quantify Li loss.

The balanced reverse reaction,



releases water even with a dry inlet, and that water can oxidize the carrier[7, ch. 4.1] (Chacko’s printed reaction omits the factor of two before the hydroxide). Equal fed CO_2 or equal mean Fe valence therefore cannot show equivalent coating–carrier interfaces, so the timing comparison measures the total effect of treatment order, interpreted with inventories and intermediate states.

2.3 Experiment 1: ordinary steam versus Ar cycling, then a timing diagnostic

Table 3 maps the three sequences; ordinary oxidation occurs every cycle, and a reset is an extra treatment.

Table 3: Sequence map for the proposed cyclic-oxide comparisons.

Sequence	Exposure	Special reset	Primary endpoint
Ordinary operation	Five-minute steam-rich or Ar post-oxidation purge; fixed block	None during the block; one after operating output is measured	Pre-reset cycle and cumulative ethylene; post-reset recovery assay
Timing diagnostic (five arms)	Two five-minute segments: wet/dry, early/late CO_2 , plus Ar reference	One, after the exposure or the fixed accumulation block	Post-reset ethylene and timing contrasts; intermediate samples for attribution
Policy blocks	Concurrent protection, delayed recovery, untreated steam, Ar at the selected duty	Managed policies: fixed cadence; others: none. Analytical reset only after the counted output	Ethylene per complete elapsed time, resource demand; terminal function separately

Ordinary-cycle comparison. Prepare one reproducible coating, measure its initial Li content, phases and loading (nominal 10 wt% is only the starting recipe), and establish stable short-cycle function under dry Ar. Then run Brody’s schedule at 700 °C and atmospheric pressure with the post-oxidation purge either steam-rich or Ar, without supplementary CO_2 ; the reducing post-ethane

purge stays dry Ar throughout, and no result addresses steam there. Set the steam fraction in Ar, flow and temperature trajectory from equipment validation and the intended heat-exchange duty; fix this composition, the initial conditioning and the averaging windows in advance, holding other cycle conditions fixed. Report the steam-minus-Ar difference in cycle-resolved and cumulative ethylene on equal initial mass and feed bases, including a terminal window *before any reset* or extra dry conditioning; it includes any carryover or reversible inhibition, so it is operating performance, not proof of permanent damage. Then apply the common reset (a fixed dry CO₂/O₂ carbonation-oxidation step; see below) once to each history and compare a common block of dry cycles with the matched Ar history: loss followed by recovery supports recoverability under that treatment, not tolerance or a unique repair mechanism. The pre-reset blocks are the untreated comparators for Experiment 2.

Timing diagnostic. If a functional or material difference warrants it, run the arms in Table 4 on separate specimens at 700 °C and atmospheric pressure. The 10 mol% steam and 5 mol% CO₂ are starting settings, not an optimum or a stand-in for nearly pure steam. Hold total inlet flow, temperature, pressure, specimen mass and elapsed exposure fixed, and match the *integrated inlet* CO₂ dose within each timing pair, with switching holdup calibrated in an empty reactor using the actual water and CO₂ signals, reporting background CO₂; uptake and release are outcomes.

Table 4: Timing and steam are crossed to separate a steam-dependent timing effect from ordinary dry-gas history. All unspecified gas is Ar.

Arm	First five minutes	Second five minutes
Wet, concurrent (P, protection)	10% steam + 5% CO ₂	Dry Ar
Wet, delayed (R, recovery)	10% steam	5% CO ₂ , dry
Dry, early CO ₂	5% CO ₂ , dry	Dry Ar
Dry, late CO ₂	Dry Ar	5% CO ₂ , dry
Ar reference	Dry Ar	Dry Ar

The dry arms test whether early CO₂ protects against a dry exposure regardless of steam. After the common reset, let q be each arm’s mean ethylene per cycle and initial catalyst mass over a fixed block, and define

$$D_{\text{steam}} = (q_{\text{wet,P}} - q_{\text{wet,R}}) - (q_{\text{dry,early}} - q_{\text{dry,late}}). \quad (10)$$

A positive wet P–R difference shows a timing benefit under steam; a positive D_{steam} adds evidence that steam changes it. Neither proves hydrolysis or Li volatilization. Also report $q_{\text{wet,P}} - q_{\text{dry,early}}$ and $q_{\text{wet,R}} - q_{\text{dry,late}}$ (steam substitution within each CO₂ schedule), and each arm’s absolute yield and its difference from Ar, since both wet arms could be equally damaged. A null result here concerns function after dilute steam, this CO₂ history and the reset, not untreated compatibility at the selected duty.

Use independently run sister specimens stopped after the first segment, the full exposure, the reset and washout, recording elapsed time and gas history. First-segment samples can locate changes present during steam; final specimens alone cannot separate steam damage from changes in the later Ar/CO₂ segment.

2.4 Common reset, function assay, and inventories

Common reset. One terminal carbonation/oxidation sequence serves the recovery comparison and all timing arms. In a pilot, choose dry-inlet CO₂/O₂ conditions and a duration long enough for

the slowest specimen to reach stable net uptake and a reproducible carbonate-sensitive spectrum, then fix it; show on this formulation that the joint endpoint limits new decarbonation while the carrier reoxidizes. Measure CO_2 , O_2 , H_2O , CO and H_2 throughout. The aim is complete treatment of the remaining carbonatable fraction (not equal carbonate if Li was lost), confirmed by the species/state check because an uptake plateau may reflect slow transport or passivation. If no shared, process-compatible reset reaches interpretable endpoints, report the recovery limit without claiming irreversibility; any further reset is applied comparably and charged to the recovery policy.

Washout and function assay. Before ethane, verify CO_2 and O_2 washout against thresholds that exclude an important change in the first pulse: Gao reports CO_2 inhibition on coated LSF in Fig. 4D (the reversible switch in Fig. 4E is a separate molten- Li_2CO_3 /ethane/ O_2 experiment),[4] and residual O_2 can change output independently of carrier oxygen. Apply the longest washout needed by any arm to all arms and check whether it alters the contrast; an effect erased by washout is not persistent protection. Then run a pilot-chosen block of dry-Ar-purged ethane/oxidation cycles long enough to separate gas carryover from retained function.

Carbon and water accounting. Directly measured ethylene is the primary outcome, not the source's C_{2+} ; also measure conversion, CO_x , other hydrocarbons and retained carbon, with a thermal blank. Include carbonate capture, release and inventory changes in the carbon balance before assigning CO_x to hydrocarbons; regeneration CO_2 alone is not a coke assay. If carbonate and coke cannot be resolved, report total CO_x and the attribution limit; the regeneration precedent faces the same accounting problem.[7, ch. 4.1] No isotope campaign is needed for the ethylene-output decision. Measure water and H_2 directly, since carbonate/hydroxide and carrier-oxygen changes make water alone an ambiguous lattice-oxygen reporter.

Lithium and spatial state. Close the Li inventory over particles, grit, reactor walls, particulates and downstream condensates by complete digestion with blanks and recovery spikes for each matrix, validating sampling and extraction recovery before aging and recording each segment's capture (earlier deposits may be released later); ordinary EDX cannot measure Li coverage. If redistribution matters, cross-sectional NanoSIMS correlated with electron microscopy could map Li-rich patches and rim-to-interior gradients, if its resolution is shown to be finer than those features and matrix-dependent ion yields, sectioning artifacts and beam-induced redistribution are controlled; intensities do not give absolute coverage, and unresolved films are outside the claim. For carrier state, use phase characterization and, if access and validation allow, ^{57}Fe Mössbauer spectra on state-preserved samples, following Gao;[4] these can show Fe-state or phase changes if reference materials and replicates resolve them above fitting uncertainty; bulk averaging can hide interfacial changes. Carbonate-sensitive vibrational spectra add a chemical indicator. Without access or validation, drop the corresponding NanoSIMS or Mössbauer mechanism claim.

Required resolution. On an assumed final-catalyst basis, 1 g with 10 wt% Li_2CO_3 holds about 18.8 mg Li, so a 0.1% change (18.8 μg) is below a 1% bulk-inventory error. Five minutes of 5% CO_2 at 100 standard cm^3/min (0 °C convention) supplies about 1.12 mmol, while converting 0.1% of the carbonate involves 1.35 μmol , about 61 ppm of the gas dose against a 50 000 ppm feed. These are resolution examples, not predicted losses; captures, absolute assays and chemical signatures must complement gas-signal subtraction. If one exposure leaves a change unresolved, repeat the timing history every cycle (with real ethane and ordinary reoxidation steps) over a fixed block and reset once at the end, fixing and reporting this cadence before assigning any loss rate; this ten-minute diagnostic exposure is not the five-minute ordinary purge duty. Raising temperature or extending steam until damage appears would test a different condition.

2.5 Experiment 2: a usable management policy and its complete-cycle value

An important ordinary-cycle penalty, recovery benefit or timing contrast justifies comparing four policies at the selected steam-rich duty: concurrent protection, delayed recovery with a demonstrated process-compatible treatment, untreated steam, and dry Ar (the last two from Experiment 1 if conditions and blocks match, otherwise matched controls). The post-ethane purge stays Ar. Fix steam dose, CO₂ circulation limit, total flow, temperature trajectory and cadence in advance; unavoidable changes in steam delivery belong to the policy. The diagnostic ratio does not transfer: steam/CO₂ = 2 in a binary purge needs one-third CO₂, which, with the source’s five-minute purge and one-minute 81.6% ethane feed and assuming equal molar flow in the purge and ethane steps, circulates about 2.04 mol CO₂ per mol ethane fed (a composition calculation, not a required dose). Keeping the steam dose instead raises total gas flow by 50% and needs a new heat and hydrodynamic comparison. If Li export is important and recovery fails, test one measured Li replenishment, counting its interruption, heating, washout and recovery; it may also change coverage, salt composition and carrier structure, and the original synthesis does not imply an online method. Let q_i be the product per cycle averaged over the whole block including aging, τ the base cycle time and h_i the average added treatment time. For equal catalyst mass and feed pulses,

$$J_i = \frac{q_i}{\tau + h_i}, \quad J_P > J_R \iff h_P < \frac{q_P}{q_R}(\tau + h_R) - \tau. \quad (11)$$

Taking $\tau = 16$ min, the unrecovered Ar-aged state as a comparator with no added time, and the published C₂₊ yield ratio as a stand-in for the per-cycle product ratio, fully restoring the Ar-aged yield from 47.24% to 53.2% permits at most

$$16 \left(\frac{0.532}{0.4724} - 1 \right) = 2.02 \text{ additional minutes per cycle} \quad (12)$$

before gas and heat are charged. A five-minute step every cycle would need 31.25% more output; complete recovery gives 12.62%. This uses the documented Ar loss, not a predicted steam loss or full-life benefit.[5] Report ethylene and C₂₊ productivity with fresh and circulating CO₂/steam, Li handling and heat per product (recycling can reduce fresh CO₂ but leaves circulation and separation burdens); a smaller Li signal alone does not show process value.

2.6 Experiment 3: predicting recoverability under a new operating history

The goal is a measured state that predicts persistent function under the recovery policy. If Li export is the leading explanation, calibrate per-cycle Li transfer under continuous protected and continuous delayed operation and relate retained Li to later ethylene output. If equal retained Li gives different function, the inventory-only model fails; test an independently measured coating-distribution or carrier-phase indicator rather than fitting a loss constant per history. Withhold one alternating protection/recovery schedule with fixed period and cadence, and predict cumulative Li transfer, cycle-resolved ethylene output, and function after its single terminal reset, which is not inserted between cycles unless it is part of the policy and charged to it. Start with an additive per-cycle loss model; add a measured state only if it explains and predicts a failure under alternation on a further block. The model must beat a carrier-state-only model and a cycle-count aging description by enough to change a recovery or purge decision. Success shows predictive use over the tested range, not that Li inventory causally dominates correlated changes.

2.7 Outcomes and decisions

The program becomes substantial if the selected steam duty changes useful function and a measured state predicts recovery.

- *No important steam-minus-Ar loss*: supports compatibility without supplementary CO₂ at the tested duty; no protection claim is needed there.
- *Loss before the reset, recovery after it*: supports recoverability under that treatment; Experiment 2 decides whether its cost is repaid.
- *No timing benefit*: drop the persistent concurrent-over-delayed claim only when the interval for the direct contrast $q_{\text{wet,P}} - q_{\text{wet,R}}$ excludes an important advantage; the interval for D_{steam} assesses whether steam modifies it. Neither shows CO₂ treatment is unnecessary; single-arm recovery bounds are separate conclusions.
- *Protection works but does not repay its complete-cycle cost against untreated steam and Ar*: keep the finding, reject the intervention.

Steam cannot explain Brody's Ar-aged decline; explaining it needs representative Ar-aged material and a separate recovery study, which must not silently replace a failed steam hypothesis.

3 Polymer ethenolysis: less fresh catalyst during reuse

Summary. Conk et al. (2024) converted polyethylene (PE) to olefins including propylene by isomerizing ethenolysis (repeated double-bond migration and metathesis with ethylene) over WO_3/SiO_2 (W/silica) mixed with $\text{Na}/\gamma\text{-Al}_2\text{O}_3$ (Na/alumina). When the retained mixture is reused over successive PE charges, output falls; one 0.4 g fresh Na/alumina addition before charge 2 restores it over three charges.[9, Fig. 4B] That rescue is our benchmark; pressure effects, initiation and regeneration, and chain-population models are also prior work.[10–12, 23, 24] The open question is whether lower ethylene partial pressure can replace part of the fresh Na/alumina, predictably. We propose to reproduce the published reuse, confirm the reuse deficit in a semibatch reactor, which pressure control requires, cross a fresh Na/alumina dose bracket with one lower ethylene pressure, predict and test a new intermediate dose, and extend the selected policy over further charges. Success would give an operating rule that needs less fresh catalyst per unit of polymer-derived carbon at specified product quality and complete cycle time, reducing catalyst preparation, solids handling and waste while preserving polymer-carbon recovery; three successful batches would not establish economic superiority. At equal output, a 0.2 g midpoint addition would use 25% less fresh Na/alumina and 16.7% less total fresh catalyst solid than the published 0.4 g addition (arithmetic, not a measurement). Selective loss of an initiating function is one possible explanation of the reuse loss, not an established result or a prerequisite for the operating rule.

3.1 Prior work

Conk and colleagues converted PE and polypropylene directly to olefins including propylene; W/silica supplies olefin metathesis and Na/alumina olefin isomerization, but these roles do not show how saturated PE first becomes reactive or which function changes during reuse. The standard experiment uses 1 g PE and 0.4 g of each solid at 320 °C, with high fresh-batch output and demonstrated reuse.[9] In the reuse benchmark the mixture is retained whole, and charge 3 receives PE alone.[9, Fig. 4B] Rescue does not show that W function is unchanged, since added Na/alumina can also alter unsaturation supply, isomerization, activation, contact, and the balance among functions.

Conk’s public dissertation abstract proposes initiation by transfer dehydrogenation with ethylene as hydrogen acceptor, rather than the paper’s cracking interpretation, and describes modified catalysts and dimethyl-ether-based regeneration; the body is embargoed, so no protocol or performance comparison can be verified.[23] Patent WO2026011187A2 discloses oxidative then dimethyl ether treatment, but its Figure 21 caption names a Na–Fe formulation and the adjoining procedures are not clearly consistent about catalyst identity, so it cannot be treated as a demonstrated regeneration benchmark for every unmodified Na/W mixture.[24]

Guironnet and Peters’ chain-population model links ethylene and reactive-chain concentrations; they describe a numerical extension to changing ethylene but display only fixed-ethylene solutions.[11] Chen, Sadow, and Peters’ final 2024 model resolves finite isomerization kinetics and predicts saturation and possible ethylene inhibition, but for a molecular catalyst with initially unsaturated chains, without deactivation or ethylene gradients, and physically restricted to relatively slow isomerization (its quasi-equilibrium limit is only a numerical check). It does not predict that lowering ethylene improves aged Na/W.[12] Wang experimentally examined pressure reduction and catalyst supply in an earlier tandem system with different materials; its pressure comparisons also changed other conditions.[10, ch. 4]

3.2 Carbon origin and catalyst demand

Gross propylene is the wrong measure because ethylene also supplies product carbon. The source’s labeled-polymer experiment shows polymer participation under its fresh-batch conditions, not a general carbon allocation for every reused inventory and pressure.[9, Fig. 4A] The accepted output Q_{PE} is the moles of carbon from supplied PE recovered in the specified accepted product over a whole sequence. For a propylene objective, count polymer-origin carbon in propylene and report higher olefins separately. Define

$$D_{\text{Na}} = \frac{M_{\text{Na/alumina,fresh}}}{Q_{\text{PE}}}, \quad D_{\text{W}} = \frac{M_{\text{W/silica,fresh}}}{Q_{\text{PE}}}, \quad J_{\text{PE}} = \frac{Q_{\text{PE}}}{T_{\text{complete}}}. \quad (13)$$

These are catalyst-solid demands, not elemental Na or W demands; report elemental Na and W inventories and retained and discarded solids separately. Complete time includes heating, reaction, cooling, gas handling, unloading or recharge, and any regeneration.

To attribute carbon, run representative sequences with ethylene at the *same enrichment throughout conditioning and every charge*, plus unlabeled reference sequences; this avoids assuming that a small labeled-PE fraction represents all chain populations. Replace the source’s methane internal standard with He, measured by calibrated GC with a thermal-conductivity detector and Ar carrier (the source’s flame-ionization detector cannot detect He), keeping separately calibrated hydrocarbon detection and synchronized sampling. First validate He recovery against known gas flows, separation from hydrogen, response in the actual gas matrix, and analytical access, and recheck reference behavior after the tracer change. Quantify initial catalyst carbon and other carbon inputs; appreciable unclassified carbon invalidates two-source attribution.

Where PE and ethylene are the only contributing carbon reservoirs and isotope discrimination is bounded, the polymer-origin fraction in product j is

$$f_{\text{PE},j} = \frac{x_E - x_j}{x_E - x_{\text{PE}}}, \quad Q_{\text{PE}} = \sum_j N_{C,j} n_j f_{\text{PE},j}, \quad (14)$$

where x denotes *carbon atom fraction* of ^{13}C , $N_{C,j}$ the number of carbon atoms per product molecule, and the sum covers accepted products collected over the sequence. Derive x_j from calibrated isotopologue distributions with natural-abundance and fragmentation corrections; the published singly labeled propylene abundance is not a carbon atom fraction. First validate isotopic response and bound fractionation or unequal enrichment within a source; do not force fractions into the interval zero to one.

Carbon collection covers gas, condensates, soluble and insoluble residue, and catalyst-bound carbon. Constant enrichment assigns retained carbon to its source but not to a particular PE charge, so decisions use cumulative sequence output; charge-resolved kinetics must state carryover. Credit collected products once, not also as disappearance of intermediate residue. A polymer-free blank bounds background reactions but does not reproduce a polymer-conditioned catalyst.

3.3 Experiment 1: reproduce reuse and bracket the fresh Na/alumina dose

With one characterized lot of clean HDPE, reproduce the catalyst preparation, one fresh time course, and the three-charge unsupplemented and once-supplemented sequences. The sealed procedure charges ethylene and methane before heating, so its nominal 15 bar ethylene is not a controlled hot partial pressure; record fill conditions, pressure convention, and actual hot pressure and temperature

history. The separate-component figure lists 10 bar where the text and supporting procedure specify 15 bar; follow the detailed method and record the discrepancy.[9, 25]

Table 5 gives the dose bracket; 0.2 g is our chosen midpoint. Each reused arm receives its fresh Na/alumina makeup before charge 2 only and keeps the whole mixture without washing or separation. The all-fresh arm, with fresh catalyst for each of three charges, is the output reference.

Table 5: Fresh catalyst solids consumed over three 1 g PE charges. The lower dose is proposed; the full once-only dose is the published comparator.

Policy	Na/alumina (g)	W/silica (g)	Total (g)
Fresh mixture for every charge	1.2	1.2	2.4
Retained, no makeup	0.4	0.4	0.8
Retained, 0.2 g before charge 2	0.6	0.4	1.0
Retained, 0.4 g before charge 2	0.8	0.4	1.2

With $f = Q_{\text{PE, reuse}}/Q_{\text{PE, allfresh}}$, total-solid productivity relative to all-fresh operation is $2f$ for the published policy and $2.4f$ for the midpoint, which gives the savings quoted in the Summary at equal output. Three doses bracket demand; they do not locate a precise optimum.

The reported restart cools and dismantles the reactor under inert conditions. A polymer-free thermal and handling cycle applied to fresh catalyst before its first PE charge tests whether interruption alone changes function (not the spent polymer environment); record solid recovery, retained carbon, exposure time, and transfer losses. Closed inert recharging, if available, is compared only after validating delivered PE mass, temperature history, and mixing; it may change more than exposure. An added-support control (same dried support at the full supplement’s solid mass) tests whether extra solids or contact explain part of the rescue; it cannot reproduce Na/alumina surface chemistry or rheology. Where possible, check agitation and particle-size sensitivity; a stirring plateau does not exclude pore-access limits. If handling dominates the loss, the outcome is a handling protocol, not evidence of intrinsic, turnover-driven deactivation.

Fix in advance the required cumulative accepted output, allowed product distribution, and complete-time ceiling, and report each charge’s output so early production cannot hide a stalled final charge.

3.4 Experiment 2: one lower ethylene pressure crossed with dose

The pressure intervention needs a semibatch reactor with continuous gas flow and product removal, whereas the benchmark is sealed batch. First establish the fresh–retained difference and the direction of Na rescue in semibatch mode; without an important reuse deficit there, stop testing pressure as a rescue.

Lower ethylene partial pressure by inert replacement at fixed total pressure, inlet molar flow, temperature, PE charge, catalyst inventory, and agitation. As a starting contrast, halve the reference inlet ethylene fraction, keep a constant carbon-free tracer fraction, and make up the balance with Ar, within the validated gas-delivery range. Measure outlet composition and actual molar flow rather than inferring reactor ethylene partial pressure from the cylinder blend, and record absolute versus gauge pressure. The source semibatch procedure’s 19 bar gauge total pressure is not interchangeable with its sealed-batch ethylene fill pressure.[25]

Match all history through charge 1 at the reference condition, then randomize retained inventories across both pressures and the Experiment 1 doses (0, 0.2, 0.4 g Na/alumina, added once before

charge 2): six conditions. Test the nonzero doses whatever the zero-dose result, because pressure and makeup can interact. Fresh mixtures at both pressures show whether a response is specific to retained catalyst. The contrast is an operating intervention on a changing mixture, not an intrinsic reaction order. The source raised semibatch output while increasing pressure and flow together; that cannot establish a pressure order, but it is substantial evidence against assuming that less ethylene must help.[9]

Compare complete charges, not rapid pressure waveforms: the published apparatus samples every 12 minutes, and gas holdup and dissolved-product release may obscure shorter responses.[25] A reference–lower–reference transient helps only if measured response times are short compared with chain evolution and catalyst change; a propylene spike from stored product is not restored production.

A useful reduced-dose policy must meet polymer-carbon attribution, full product collection, product specification, and the complete-time ceiling. Compare each reduced dose at both pressures with the full-dose, reference-pressure policy, reporting the pressure contrast within each dose. A null result at zero dose rules out only pressure rescue without makeup; a broader saving claim also needs the reduced nonzero dose. If the fresh comparison supports it, test lower ethylene from the start of all three charges as a separate comparison (its charge-1 history differs from the six-condition cross); a fixed condition that performs as well as a history-dependent switch is preferred. Count inert-gas supply, separation, and circulation; lower partial pressure at unchanged total pressure does not imply lower compression demand or stoichiometric ethylene use.

The opposite outcome is equally plausible: ethylene takes part in metathesis, may accept hydrogen during initiation, and changes swelling and gas–polymer transport, so lowering it can slow one function more than it relieves another, shift selectivity, or reduce contact. Drop the partial-substitution claim if the crossed comparisons exclude an important saving that meets the output, specification, and time requirements. Either way, the result does not identify which function decayed, and a pressure library is not justified before an apparent improvement appears.

3.5 Chain-state evidence and selective-initiation claims

Product trajectories are tracked with full molecular-weight distributions, recovered fractions, and, where measurable, unsaturation. Endpoint olefin content reflects formation and consumption together, so low unsaturation does not prove slow initiation, nor does a bulk C=C fraction show how many chains carry reactive sites; heating alone changes molecular weight in the source’s component experiments, so mean molecular weight alone cannot assign mechanism.[9] Because depressurization and two to three hours of cooling allowed isomerization and fragment recombination in Wang’s tandem system,[10, ch. 4, pp. 72–74] compare the fastest validated product workup with a deliberately delayed workup at the same conversion window. Solvent-soluble material cannot represent carbon retained on inaccessible solids.

If pressure gives a real saving, a matched unsaturated PE and its hydrogenated counterpart can test whether feed state changes the benefit, with chain-length and branching distributions, workup residues, and metal contamination controlled and a prediction from the measured Na/W response, not the optimum of Chen et al.’s molecular Ru-catalyst model. It does not isolate initiation, because unsaturation also changes chain distribution and contact. A component-specific claim needs probes shown to be selective on the separate catalysts and supports, since small olefins may reach sites that polymer cannot or activate the material; otherwise, keep an empirical relation among history, pressure, and fresh-material demand.

3.6 Experiment 3: predict a new dose, then test whether the saving persists

Build the simplest predictor of accepted output at both pressures from the crossed brackets and complete time courses, keeping any pressure–dose interaction: start with interpolation over the measured makeup range and its uncertainty; do not fit separate unobserved initiation and metathesis capacities to one product trace. If output is not regular enough in dose to interpolate, choose among tested policies.

Validation. Choose a new intermediate dose (not 0, 0.2 or 0.4 g) only if the predictor says it meets the output and time criteria with a resolvable saving. Test it at *both* pressures on three-charge sequences with matched charge-1 histories, added before charge 2 only, against the best tested fixed-pressure policy and the published full-supplement schedule in the same reactor mode. It succeeds if the lower confidence bound on accepted output meets the requirement, specification and time are met, and fresh-material demand is lower after attribution uncertainty. Pressure enables the saving only if the lower-pressure policy passes and the same dose at reference pressure fails an acceptance criterion beyond its uncertainty. If both pass, report an adequate lower dose and any separately resolved pressure benefit, without claiming pressure was needed; an unresolved reference result leaves this open. No minimum over unmeasured doses is identified.

Extension. To test whether the saving only postpones replacement, run the selected policy and its comparator through further 1 g PE charges toward a common cumulative accepted-output target, within maximum complete time and charge count, all fixed in advance with a per-charge replacement trigger (accepted output, quality, charge-time ceiling). At the trigger, remove the whole catalyst–residue mixture under the validated handling protocol, quantifying its solids and unprocessed carbon; add 0.4 g fresh Na/alumina and 0.4 g fresh W/silica; and restart the assigned pressure schedule, with once-only makeup before the new mixture’s second charge and the common reference first charge where validation used it. No selective W replacement or unvalidated regeneration. Count failed charges and all removal, recharge, and reheating time; residues stay in the carbon and waste accounts and are not recovered product. Credit the common target as output and report excess separately; a policy that misses the target gets no extrapolated credit. The conclusion covers only this extension and replacement rule; a shorter service interval pays its measured replacement cost.

The scientific payoff grows if a chain or catalyst-state descriptor predicts a new history better than charge count alone and so changes a makeup decision; otherwise a reproducible fixed-pressure saving is still a useful operating result. If no validated intervention lowers demand at accepted output and time, drop the improvement claim. Neither a negative result nor fresh Na/alumina rescue identifies a unique molecular cause of reuse loss.

4 Promoted Ag: when does dilute nickel add useful ethylene oxide output?

Summary. Promoted Ag epoxidation catalysts must favor ethylene oxide (EO) over combustion at an economic rate, and they lose performance in service. Ni-dependent selectivity retention on a fully promoted catalyst is already reported: Kemp’s 1995 Ni-containing Ag/Cs/Re/S catalyst lost 4.3 selectivity points over 50 days of accelerated aging versus 6.1 for its Ni-free parent, though the pair also differed in Cs content and post-treatment.[13] Jalil and colleagues showed that dilute Ni suppresses combustion more than EO formation on supported Ag; their main materials lack the full Cs/Re promoter system.[14] Chloride operation could equally explain a retention difference: measured and modeled chloride responses and prior lower-chloride disclosures show that it alone can change selectivity and output.[15, 16, 26, 27] The original contribution is therefore narrow: a composition- and processing-controlled, Ni-dependent relationship that predicts when changing chloride operation can substitute for Ni, or when a useful Ni advantage survives it; better controls alone do not establish it, and a practical benefit needs no unique microscopic explanation. The program compares ordinary drift of Ni-treated, sham-processed and untreated material over a fixed horizon, including the best tested chloride policy for each (Stages A–B), tests any difference against a calibrated chloride-response baseline (Stage C), and, if the loss is partly recoverable, predicts when a chloride recovery step pays off (Stage D). Success would deliver a reproducible formulation/policy comparison and a validated prediction of where any Ni advantage ends. As an illustration of the stakes, not a predicted improvement: if EO and complete combustion are the only carbon outlets, raising ethylene selectivity from 0.90 to 0.91 at fixed EO output lowers reaction-level ethylene use by 1.10%, oxygen use by 4.40% and CO₂ formation by 10.99%.

4.1 Prior work and the remaining question

The Summary’s calculation uses $1/S$ moles of ethylene, $0.5 + 3(1/S - 1)$ moles of oxygen and $2(1/S - 1)$ moles of CO₂ per mole of EO at fractional ethylene selectivity S . It excludes recycle, separation, utilities and catalyst manufacture, whose costs a measured improvement must also survive.

Kemp added Ni after preparation of the Ag/Cs/Re/S catalyst; lithium-containing precursors were also used. Selectivity was measured at 40% oxygen conversion after reoptimizing. Both catalysts started within a reported 86.0–86.5% range, and the bulk Ni:Ag atomic ratio was about 1:835. A solvent-treated, Cs-matched Ni-free control started worse and was not aged. The result is a difference between losses from separate starting values, not an exactly measured final-selectivity difference or a demonstrated lifetime extension, and it neither isolates the Ni effect nor reports useful integrated output.[13]

In Jalil’s work, organochloride raised selectivity further. Processing controls, model surfaces and Ni spectroscopy support a Ni-associated effect, but nominal loading does not count working surface Ni. In the supplementary chloride-response trace, the EO rate on NiAg first rises and then declines substantially: an evolving Ni–chloride interaction, not a stationary, durable output advantage on the proposed formulation.[14, 28]

Hwang and colleagues link the distinctive Cs/Re/chloride response to promoter architecture and propose that otherwise unselective oxygen becomes an additional epoxidizing intermediate. This motivates asking whether Ni is compatible with that architecture, but is not direct evidence that Ni supplies oxygen to Re or preserves their contact.[29]

Chloride operation could equally explain a retention difference. Adjusting retained chlorine (deposited from alkyl-chloride modifiers, removed by competing processes including alkanes such as ethane) can trade selectivity against useful EO rate; the optimum depends on material and conditions, and no universal coverage or modifier setting follows.[15] Iyer and Bhan already predicted reactor output from independently measured oxidation, chlorine-exchange and support-mediated EO degradation kinetics, and Santos and colleagues showed that temperature, oxygen, water and CO₂ change the chloride response.[15, 16] Prior disclosures describe lower-chloride aging policies (28-day tests, then extrapolation) and materials with a more stable optimum chloride requirement that hold work rate by adjusting temperature. Neither a crossover extrapolated well beyond the 28-day data nor a stable chloride optimum proves stable activity.[26, 27]

Because selectivity alone can reward an almost inactive catalyst, the endpoint is sustained EO output. The program does not assume that Ni preserves a particular oxygen species or Ag-Re arrangement, and a mechanistic claim needs a discriminator beyond a selectivity curve.

4.2 Stage A: validate the materials and define a fair output comparison

Materials. Use an existing, reproducible Cs/Re-promoted Ag/ α -Al₂O₃ reference and its validated epoxidation reactor; access to both is an entry requirement. Hwang’s Ag/support-area dependence favors keeping this working architecture over moving the promoters onto a dilute colloidal-Ag support.[29] From each parent preparation, make untreated, sham-processed and Ni-treated materials. The first Ni route is Kemp’s post-impregnation with a nickel-nitrate/amine-containing solution and staged drying, adapted after a small compatibility check. Start near a bulk Ni:Ag atomic ratio of 1:800, an illustrative choice close to the historical loading, rather than assuming that Jalil’s 1:200 optimum transfers. The sham receives every processing step except the Ni precursor; document any unavoidable solution-composition difference. Measure Ag, Ni, Cs, Re and other promoter contents and Ag size distributions; matched bulk contents do not establish identical promoter placement.

The sham is the causal comparator for the Ni-addition step. The untreated reference detects processing damage and represents ordinary operation of the starting catalyst; because a processing penalty can appear after the fresh check, it also runs over the full horizon. If the sham loses the reference’s useful response, allow one correction for an identified preparation problem, then stop this route if the problem remains. A Ni/sham retention difference without a useful advantage over untreated operation is processing dependent, not an improvement of the starting catalyst. The Kemp route does not establish isolated Ni in Ag; comparing unpromoted Ag and Ni-Ag made the same way is needed only to attribute the result to the Ni function Jalil reported on fresh catalysts.

Measurement and operating conditions. Before assessing small selectivity differences, commission temperature uniformity, transport sensitivity and feed-to-analyzer response. Close the carbon balance against a calibrated flow tracer, counting ethane fed to remove chlorine as a carbon source and separating net formation from outlet concentration when CO₂ or EO is cofed. Report selectivity from consumed ethylene, because $r_{EO}/(r_{EO} + r_{CO_2}/2)$ holds only under its carbon-source and product assumptions; unresolved cofeed contributions limit interpretation.

Condition all three materials by one protocol rather than forcing equal initial selectivity by different treatments. Map a small overlapping bracket of chloride settings with returns to a reference setting. Hold temperature, pressure, ethylene, oxygen, ethane and flow common for Ni and sham, and find a useful untreated setting within the same operating and tuning limits. Specify absolute pressure and the actual chloride feed, not only a nominal chloriding index. Set dwell times from observed

rates and gas holdup, up to a fixed maximum; if reference returns remain history dependent, report age-specific trajectories rather than a composite steady curve. These data answer two questions: does Ni change EO and combustion rates at common conditions, and which material gives more accepted EO at its own chosen setting? A shifted chloride optimum can answer the second without altered oxygen chemistry. If both materials pass validation, no difference between the fresh materials does not stop the program, since retention may differ later.

Accepted output and cost boundary. Before the ordinary-operation comparison, fix the output requirements (ethylene-conversion interval, selectivity floor, minimum EO work rate, and temperature and feed limits) from the reproduced reference and the intended operating decision, not from whichever endpoint favors Ni, and fix a finite horizon H that starts with initial conditioning. For policy p ,

$$P_p(H) = \frac{Q_p(H)}{m_{\text{Ag},0}H}, \quad Q_p(H) = \int_0^H \dot{n}_{\text{EO},p}^{\text{accepted}}(t) dt. \quad (15)$$

EO is accepted only if it meets the product criteria, judged in windows compatible with analyzer response; off-spec product and downtime count as zero. Conversion, selectivity and work-rate requirements apply in production windows and a return-assessment window, both fixed in advance; temperature and feed limits apply throughout. Planned conditioning, recovery and return intervals may produce little or nothing but have fixed maximum durations and a fixed return deadline. A policy that fails a production window or misses the deadline is inadmissible, and a low-output interval cannot be relabeled afterward as planned recovery. The clock includes chloride searches that recur in the policy. Also report unfiltered net EO, failure times, temperature history, and output per total catalyst mass and bed volume, since Ag normalization can hide dilution.

For each admissible policy p against a named tested comparator b , the preparation-cost boundary per bed campaign over H , at matched initial Ag inventory and with actual bed masses and volumes reported, is

$$C_{\text{prep},p} - C_{\text{prep},b} \leq v_{\text{EO}} [Q_p(H) - Q_b(H)] - \sum_k v_k [U_{k,p}(H) - U_{k,b}(H)] - v_L [L_p(H) - L_b(H)], \quad (16)$$

where v_{EO} is the value per mole of accepted EO, U_k are resource or disposal quantities (measured feed and modifier, utilities on a stated basis, valued emissions or waste) with unit costs v_k , and L is operating handling labor at v_L per labor-hour. Preparation cost is not counted again as an operating resource, and downtime, already charged through output, is not automatically labor. Without justified valuations, report the symbolic boundary and physical differences, not a monetary saving.

4.3 Stage B: compare ordinary drift and operating policies

Within each preparation block, assign independent charges to the Ni/sham common-setting pair and to the useful policies from Stage A, including at least one independently replicated untreated policy, all starting with the same conditioning and running for H . A common-setting arm may double as a policy arm only if its entire history is identical. The common-setting pair measures the effect of Ni addition under that exposure; the policy arms compare what an operator could use, and the two are not interchangeable.

Fix the untreated policy before aging from its own chloride response: one lower-chloride setting if it meets the requirements, otherwise a credible reference setting or scheduled adjustment, not a

fitted percentage of another catalyst’s optimum. Ni and sham get the same operating limits and tuning effort. If holding accepted conversion needs a different flow or temperature, that change and its consequences belong to the policy. The untreated policy is required even if its fresh response matches the sham’s; it needs no full chloride or recovery factorial, but the best tested untreated policy appears explicitly in the rankings.

Compare cumulative EO and reactant use over H , not endpoint selectivity: a lower-chloride policy may pay a continuing selectivity penalty, not a one-time startup cost, and a temperature controller can hide activity loss unless its trajectory is kept. Higher selectivity with lower work rate or higher temperature remains a tradeoff. No accelerated exposure is needed. If neither drift nor a useful material or policy difference is resolved within H , end the program at that preparation and operating scope; a severe chloride pulse or an extrapolated crossover cannot substitute.

If an important loss develops, compare one mild recovery step with continuation on matched-age beds of both materials that share the common-setting history. Take the recovery chloride setting from the established bracket, hold other feeds fixed, then return both beds to the common reference setting for an observation window fixed in advance. Follow the recovery and continuation arms separately afterward, since treatment may change later drift. This tests recoverable conditioning under that intervention, not general regeneration. A failed recovery shows persistence over the tested duration, not irreversibility.

4.4 Stage C: test whether the difference exceeds the known chloride response

The baseline is each material’s measured chloride/product response, informed by established chlorine-exchange and reactor models and calibrated on the actual formulation and feed range, including response times and, where conversion requires it, axial product gradients (less combustion makes less water and CO_2 in the bed, so equal inlet products need not mean equal local states).[15, 16] An uncalibrated literature model is not a fair baseline.

Compare the EO– CO_2 trajectory during recovery with independently measured steady or matched-age chloride-response curves. A trajectory that follows the curves can still show a useful Ni-dependent recovery speed, since the curves do not set how fast the bed moves along them. Departure from a whole-bed curve can reflect axial mixtures of chlorine states, temperature or product gradients, so it does not identify promoter relocation. Add reproducible rate memory (a slowly changing state that makes the current rate depend on past chloride history, not a count of selective oxygen sites) only if the calibrated baseline misses by more than its prediction uncertainty and an important difference remains.

Use at most one further discriminator, chosen by the result. If less self-generated CO_2 seems to explain the Ni benefit, compare the pair under a common, externally controlled product burden and predict an untested burden, accounting for bed gradients. If the loss survives reconditioning and an inventory change is plausible, examine retained elements and Ag morphology on sister samples; unchanged total Re does not exclude redistribution, and a changed mean particle size need not explain all lost function. EO cofeed measures only net product sensitivity, and contact-time curvature cannot identify a secondary combustion flux.

A working-state assay is justified only if it separates the remaining alternatives and can detect the minority population invoked; an unchanged element-average spectrum cannot exclude it when its abundance is unconstrained. Chlorine stripping gives an operational inventory, and trifluoroethanol titration is destructive and estimator dependent: Iyer and Bhan treat total uptake as an upper-bound site estimate because inactive adsorption is possible.[30] Neither gives an exact active-site count

on Ni-containing material without new validation. Oxygen-isotope return measurements and a Ni-by-Re composition series are optional extensions. Isotope return needs a justified link between labeled and returning oxygen, a complete balance and a resolvable signal; neither high selectivity nor undetected gas-phase return proves Ni-to-Re oxygen transfer.

4.5 Stage D: predict when a chloride recovery step pays off

If drift is partly recoverable, fit, for Ni and sham, the matched-age recovery and continuation traces with the simplest time-dependent description consistent with both EO and CO₂ rates. A single relaxation time is allowed only if its approach and return are resolved and repeatable. Check it on a second training duration, then choose a withheld duration within the tested range. The prediction must not rely on rates identifying surface coverage.

Over a common remaining horizon H_r , the recovery benefit relative to continuation is

$$B_m(t_R; H_r) = \int_0^{H_r} \left[\dot{n}_{\text{EO},m,R}^{\text{accepted}}(t; t_R) - \dot{n}_{\text{EO},m,C}^{\text{accepted}}(t) \right] dt, \quad (17)$$

where m is Ni or sham, t_R is the recovery duration, and t is measured from the intervention at operating age a (whereas t in Eq. (15) starts at conditioning), so $H_r = H - a$. The integral includes recovery and return time, not just recovered peak activity. Resources are compared over the same interval. The model predicts the sign of B_m and the EO and CO₂ trajectories for both materials.

B_m compares policies within one material. To rank across materials, convert it to absolute output,

$$Q_{m,R}(H) = Q_{m,C}(H) + B_m(t_R; H - a), \quad m \in \{\text{Ni, sham}\}, \quad (18)$$

using that material's continuation baseline with identical pre-intervention history, clock and inventory basis. Rank *all admissible tested* policies (Ni ordinary and recovery, sham ordinary and recovery, untreated ordinary) at the same H under the same acceptance and resource rules, with inventory-normalized outputs and the cost boundary on the matched-inventory basis. A larger within-material benefit does not decide absolute output. A comparison against ordinary Ni alone supports only that narrower conclusion; the ranking covers the tested policies, not an unmeasured optimum.

Recovery learned under the common-setting history cannot be assumed to transfer to another policy history; that would need new validation. The withheld test repeats recovery and matched continuation of both materials on independent preparation blocks, with return setting and acceptance windows fixed in advance, and checks both absolute outputs and the ranking against the ordinary policies, including untreated. Do not retune chloride afterward. Sham recovery beating ordinary Ni does not remove a Ni advantage if Ni recovery is better, so the Ni recovery policy stays in the ranking.

Success requires an observed benefit or limit within H_r that changes the operating decision, predicted accurately enough to resolve the smallest important difference. A calibrated chloride-response model that already predicts the result supports an operating choice but ends the claim of an additional Ni-specific mechanism; a Ni-dependent rate-memory term must beat it on withheld prediction and identify a useful preparation or operating dependence. A periodic-service model needs repeated returns to the same productive state and measured recurrence costs, not one successful recovery.

If the ordinary difference is persistent rather than recoverable, replace this stage with one product burden or exposure duration, fixed in advance, that the remaining explanation predicts will keep or remove the Ni advantage; do not pursue the recoverable and persistent branches in parallel. A result that only reproduces historical Ni retention, disappears under chloride adjustment without a

further important relationship, or needs an open-ended synthesis campaign should remain a bounded negative or confirmatory study and not be expanded.

5 Reserve ideas and withdrawn claims

The six ideas below are ranked for possible reconsideration, not proposed as six further full campaigns. Each entry gives the question, the prior work it builds on, and the next step. Existing equipment may change the order in which a small test is worthwhile.

1. Tungsten coordination and retention in sugar conversion: exploratory hold. *Question:* can a controllable anchoring or feed variable link productive sugar coordination, W export and useful glycol output? *Prior work:* Ti–O–W stabilization and catalyst reuse already have strong precedent. In Li’s study, the supported-Ni control without intentionally added W (Ni/M) gives 29.5% ethylene-glycol yield with preleachate, which must be compared with 27.1% for Ni/M alone; 29.5% is not a leachate-derived yield.[31, 32] *Next step:* if the idea is reopened, first reproduce a retained and a soluble-W reference with hot export and carbon balances. Transferred activity can show exported function but cannot uniquely assign all upstream turnover. The idea stays on hold: a chemically grounded intervention addressing an important limitation of the reference is still missing.

2. Product-rich liquid Ti-zeolite epoxidation: conditional model-transfer test. *Question:* does a calibrated reaction network predict an independent product-accumulation trajectory, including peroxide loss and epoxide ring opening? *Prior work:* product-induced solvent changes already have direct spectroscopic precedent, and related Ti-zeolite secondary kinetics contain unresolved behavior.[33, 34] *Next step:* validate component kinetics in the chosen liquid catalyst/solvent system before fixing the prediction; a vapor-fed model is not a validated liquid baseline. If the established network predicts useful output adequately, drop the claim of an additional mechanism. A residual alone does not identify solvent reorganization, fouling or Ti loss.

3. Zirconia styrene self-cleaning: limited chemical-balance pilot. *Question:* does ethylbenzene exposure remove inhibiting water from Zr–O acid–base pairs, and does the exported oxygen track recovered function? *Prior work:* this hydrocarbon-assisted water removal is already proposed, but the cleaning products were not identified in the source’s dilute-feed regime.[35] *Next step:* first establish useful net output with measured moisture and relevant styrene/hydrogen burdens. Then cross water history with ethylbenzene exposure to relate oxygen export to recovered function. Control oxygen exchange and perturbation by the activity assay before isotope attribution. Expand only if the chemistry predicts a useful feed or removal policy; omitting steam alone does not establish process savings.

4. Does an acid-site assay change subsequent zeolite aging? Conditional protocol study. *Question:* compared with uninterrupted aging and a matched thermal/water sham, do repeated NH_3 /water assays change the later aging of protonic chabazite (H-CHA)? *Prior work:* repeated assays are part of the relevant published protocol. A Cu/SSZ-13 aging-order comparison assayed both arms, so it does not compare assay with no assay.[36, 37] *Next step:* measure function on separate matched specimens before terminal titration; a common terminal assay then compares recoverable capacity. A useful result would predict later function under a withheld assay spacing. Immediate uptake recovery is not enough, and a null schedule effect cannot by itself determine the lifetime of an unseen Al intermediate.

5. Wet methane oxidation after NO removal: application-dependent diagnostic. *Question:* does the sulfur-tolerance advantage of Pd/SSZ-13 treated by ST-CO-N₂-O₂ over its ST-CO-O₂ counterpart (which lacks an extra N₂ treatment between CO and O₂) survive removing NO at matched water, sulfur history, temperature and precious-metal inventory? *Prior work:* the reported sulfur interval includes NO, and history-dependent NO promotion on other Pd catalysts is already known.[38, 39] *Next step:* run the small crossed NO/sulfur test on both variants only where a real low-NO, sulfur-containing duty could change the material choice. Without such an application, another NO response is a diagnostic, not a full program.

6. Ga-based propane dehydrogenation regeneration: conditional cycle study. *Question:* does a measured restoration endpoint predict a shorter useful regeneration for a withheld spent history? *Prior work:* redox-related deactivation and shortened oxygen treatments are already reported.[40, 41] Oxygen consumption minus CO₂ alone is not a selective Ga-redox inventory, because oxidation of retained hydrogen and other oxygen-containing products also enters the balance. *Next step:* calibrate that balance before assigning sites. Then compare complete-cycle propylene output, gas use and heat duty with the best simple regeneration policy. Correcting an assay does not by itself establish a better catalyst cycle.

Withdrawn claims

Several attractive shortcuts do not survive scrutiny. A faster outlet-water transient does not uniquely establish lower active-site water exposure (Section 1). Fresh-Na rescue does not identify selective loss of initiation (Section 3). Ni compatibility and retention on promoted Ag are already prior art (Section 4). The chapters replace these claims with narrower functional and predictive questions; none of the corrections invalidates its catalytic field.

For tungsten, the premise that mannose forms an inactive reservoir is also withdrawn. Published hot mannose feeds produce glycolaldehyde, although epimerization complicates assignment of the direct cleavage route.[42] This does not exclude an inactive fraction, but such a reservoir cannot be assumed.

This section thus separates missing evidence from demonstrated contradiction. An incomplete idea can be reconsidered after its specific missing test; an invalid inference should not be kept as the default mechanism.

References

- [1] Wei Fang, Chengtao Wang, Zhiqiang Liu, et al. Physical mixing of a catalyst and a hydrophobic polymer promotes CO hydrogenation through dehydration. *Science*, 377:406–410, 2022. doi: 10.1126/science.abo0356. URL <https://doi.org/10.1126/science.abo0356>.
- [2] Hangjie Li, Wei Fang, Ling-Xiang Wang, et al. Physical regulation of copper catalyst with a hydrophobic promoter for enhancing CO₂ hydrogenation to methanol. *The Innovation*, 4: 100445, 2023. doi: 10.1016/j.xinn.2023.100445. URL <https://doi.org/10.1016/j.xinn.2023.100445>.
- [3] Wei Fang, Yu Hui, Yuhui Chen, et al. Hydrophobic promoter in physical proximity stabilizes cobalt catalysts for durable Fischer–Tropsch synthesis. *Nature Communications*, 17:9608, 2026. doi: 10.1038/s41467-026-76571-8. URL <https://doi.org/10.1038/s41467-026-76571-8>.
- [4] Yunfei Gao, Xijun Wang, Junchen Liu, Chuande Huang, Kun Zhao, Zengli Zhao, Xiaodong Wang, and Fanxing Li. A molten carbonate shell modified perovskite redox catalyst for anaerobic oxidative dehydrogenation of ethane. *Science Advances*, 6:eaa9339, 2020. doi: 10.1126/sciadv.aaz9339. URL <https://doi.org/10.1126/sciadv.aaz9339>.
- [5] Leo Brody, Luke Neal, Junchen Liu, and Fanxing Li. Autothermal chemical looping oxidative dehydrogenation of ethane: Redox catalyst performance, longevity, and process analysis. *Energy & Fuels*, 36:9736–9744, 2022. doi: 10.1021/acs.energyfuels.2c01293. URL <https://doi.org/10.1021/acs.energyfuels.2c01293>.
- [6] Timothy A. Barckholtz, Heather Elsen, Patricia H. Kalamaras, Gabor Kiss, Jon Rosen, Dario Bove, Emilio Audasso, and Barbara Bosio. Experimental and modeling investigation of CO₃²⁻/OH⁻ equilibrium effects on molten carbonate fuel cell performance in carbon capture applications. *Frontiers in Energy Research*, 9:669761, 2021. doi: 10.3389/fenrg.2021.669761. URL <https://doi.org/10.3389/fenrg.2021.669761>.
- [7] Dennis Chacko. *Molten-Salt-promoted Redox Oxide Catalysts for Chemical-looping Oxidative Dehydrogenation of Ethane*. PhD thesis, North Carolina State University, 2025. URL <https://repository.lib.ncsu.edu/items/f251e28c-eff7-48f8-89a5-4d2ed038155c>.
- [8] Sonia Fereres, Cristina Prieto, Pau Giménez-Gavarrell, Alfonso Rodríguez, Pedro Enrique Sánchez-Jiménez, and Luis A. Pérez-Maqueda. Molten carbonate salts for advanced solar thermal energy power plants: Cover gas effect on fluid thermal stability. *Solar Energy Materials and Solar Cells*, 188:119–126, 2018. doi: 10.1016/j.solmat.2018.08.028. URL <https://doi.org/10.1016/j.solmat.2018.08.028>.
- [9] Richard J. Conk, Jules F. Stahler, Jake X. Shi, Ji Yang, Natalie G. Lefton, John N. Brunn, Alexis T. Bell, and John F. Hartwig. Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts. *Science*, 385:1322–1327, 2024. doi: 10.1126/science.adq7316. URL <https://doi.org/10.1126/science.adq7316>.
- [10] Nicholas Mason Wang. *Homogeneous Flow Catalysis Strategies for Lab-Scale Kinetic Investigation and Reaction Development*. PhD thesis, University of Illinois Urbana-Champaign, 2022. URL <https://www.ideals.illinois.edu/items/126888>.

- [11] Damien Guironnet and Baron Peters. Tandem catalysts for polyethylene upcycling: A simple kinetic model. *The Journal of Physical Chemistry A*, 124:3935–3942, 2020. doi: 10.1021/acs.jpca.0c01363. URL <https://doi.org/10.1021/acs.jpca.0c01363>.
- [12] Ziqiu Chen, Aaron D. Sadow, and Baron Peters. A microkinetic model for isomerizing ethenolysis. *ACS Catalysis*, 14:6339–6348, 2024. doi: 10.1021/acscatal.4c00465. URL <https://doi.org/10.1021/acscatal.4c00465>.
- [13] Richard A. Kemp. Process for preparing ethylene oxide. US Patent US5380885A, 1995. URL <https://patents.google.com/patent/US5380885A/en>. Catalysts A–C and Table III.
- [14] Anika Jalil, Elizabeth E. Happel, Laura Cramer, Adrian Hunt, Adam S. Hoffman, Iradwikanari Waluyo, Matthew M. Montemore, Phillip Christopher, and E. Charles H. Sykes. Nickel promotes selective ethylene epoxidation on silver. *Science*, 387:869–873, 2025. doi: 10.1126/science.adt1213. URL <https://doi.org/10.1126/science.adt1213>.
- [15] Krishna R. Iyer and Aditya Bhan. Interdependencies among ethylene oxidation and chlorine moderation catalytic cycles over promoted Ag/ α -Al₂O₃ catalysts. *ACS Catalysis*, 11:14864–14876, 2021. doi: 10.1021/acscatal.1c03493. URL <https://doi.org/10.1021/acscatal.1c03493>.
- [16] Vera P. Santos, Anthony Plauck, Jake Gold, Paulami Majumdar, Mark H. McAdon, and Ted Calverley. The complex chlorination effects on high selectivity industrial EO catalysts: Dynamic interplay between catalyst composition and process conditions. *ACS Catalysis*, 14:10839–10852, 2024. doi: 10.1021/acscatal.4c01764. URL <https://doi.org/10.1021/acscatal.4c01764>.
- [17] M. Wolf, N. Fischer, and M. Claeys. Capturing the interconnectivity of water-induced oxidation and sintering of cobalt nanoparticles during the Fischer–Tropsch synthesis in situ. *Journal of Catalysis*, 374:199–207, 2019. doi: 10.1016/j.jcat.2019.04.030. URL <https://doi.org/10.1016/j.jcat.2019.04.030>.
- [18] K. F. Hanssen, E. A. Blekkan, D. Schanke, and A. Holmen. Transient and steady-state studies of the effect of water on cobalt Fischer–Tropsch catalysts. In G. F. Froment and K. C. Waugh, editors, *Dynamics of Surfaces and Reaction Kinetics in Heterogeneous Catalysis*, volume 109 of *Studies in Surface Science and Catalysis*, pages 193–202. Elsevier, 1997. doi: 10.1016/S0167-2991(97)80407-7. URL [https://doi.org/10.1016/S0167-2991\(97\)80407-7](https://doi.org/10.1016/S0167-2991(97)80407-7).
- [19] James Paterson, Mark Peacock, Hendrik Van Rensburg, and Zhuoran Xu. Fischer–Tropsch cobalt activations: The role of water on catalyst reduction. *Catalysis Today*, 430:114559, 2024. doi: 10.1016/j.cattod.2024.114559. URL <https://doi.org/10.1016/j.cattod.2024.114559>.
- [20] Jinglin Yang, Xuejin Fang, Yuebing Xu, and Xiaohao Liu. Investigation of the deactivation behavior of Co catalysts in Fischer–Tropsch synthesis using encapsulated Co nanoparticles with controlled SiO₂ shell layer thickness. *Catalysis Science & Technology*, 10:1182–1192, 2020. doi: 10.1039/C9CY02557J. URL <https://doi.org/10.1039/C9CY02557J>.
- [21] Sanghamitra Sengupta, Alexander J. Klaver, Qingyuan Zheng, Marco de Jong, Robert L. C. Voeten, Lynn F. Gladden, and G. Leendert Bezemer. Hydrophilic or hydrophobic? how byproducts change the water affinity of Fischer–Tropsch catalysts. *JACS Au*, 5:5665–5675, 2025. doi: 10.1021/jacsau.5c01157. URL <https://doi.org/10.1021/jacsau.5c01157>.

- [22] Stefano Brandani and Enzo Mangano. Direct measurement of the mass transport coefficient of water in silica-gel using the zero length column technique. *Energy*, 239:121945, 2022. doi: 10.1016/j.energy.2021.121945. URL <https://doi.org/10.1016/j.energy.2021.121945>.
- [23] Richard James Conk. *Valorization of Polyolefins through Olefin-Intermediate Processes*. PhD thesis, University of California, Berkeley, 2025. URL <https://escholarship.org/uc/item/7v07b7c8>. Public abstract only; dissertation body embargoed until September 30, 2027.
- [24] John F. Hartwig, Richard J. Conk, and Alexis T. Bell. Conversion of polyolefins. Patent application WO2026011187A2, 2026. URL <https://patents.google.com/patent/WO2026011187A2/en>. Regeneration disclosure, paragraphs 0172–0177 and Figure 21 caption.
- [25] Richard J. Conk, Jules F. Stahler, Jake X. Shi, et al. Supporting information for: Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts. Supplement to Science, DOI: 10.1126/science.adq7316, 2024. URL <https://doi.org/10.1126/science.adq7316>. General methods, pp. S6–S8; semibatch methods, pp. S16–S17.
- [26] Wen-Sheng Lee, Mark H. McAdon, Edward M. Calverley, and Vera Santos Castro. Process for reducing the aging-related deactivation of high selectivity ethylene oxide catalysts. Patent application US20240279193A1, 2024. URL <https://patents.google.com/patent/US20240279193A1/en>. Example 2.
- [27] John Robert Lockemeyer and Randall Clayton Yeates. Process for the production of ethylene oxide. Patent application US20240025869A1, 2024. URL <https://patents.google.com/patent/US20240025869A1/en>. Examples 1–7.
- [28] Anika Jalil, Elizabeth E. Happel, Laura Cramer, et al. Supplementary materials for: Nickel promotes selective ethylene epoxidation on silver. Supplement to Science, DOI: 10.1126/science.adt1213, 2025. URL <https://doi.org/10.1126/science.adt1213>. Preparation and conditioning methods; Figures S14–S16, S21 and Table S5.
- [29] Andrew Hwang, Carlos Lizandara-Pueyo, Andrey Karpov, and Enrique Iglesia. Mechanism and site requirements for dioxygenase channels in selective ethene oxidation on promoted Ag catalysts. *Journal of Catalysis*, 461:117021, 2026. doi: 10.1016/j.jcat.2026.117021. URL <https://doi.org/10.1016/j.jcat.2026.117021>.
- [30] Krishna R. Iyer and Aditya Bhan. Chemical titration of promoted Ag/ α -Al₂O₃ ethylene epoxidation catalysts. *ChemCatChem*, 15:e202300329, 2023. doi: 10.1002/cctc.202300329. URL <https://doi.org/10.1002/cctc.202300329>.
- [31] Naixu Li, Xu Liu, Jiancheng Zhou, Quanhong Ma, Maochang Liu, and Wenshuai Chen. Enhanced Ni/W/Ti catalyst stability from Ti–O–W linkage for effective conversion of cellulose into ethylene glycol. *ACS Sustainable Chemistry & Engineering*, 8:9650–9659, 2020. doi: 10.1021/acssuschemeng.0c00836. URL <https://doi.org/10.1021/acssuschemeng.0c00836>.
- [32] Naixu Li, Xu Liu, Jiancheng Zhou, et al. Supporting information for: Enhanced Ni/W/Ti catalyst stability from Ti–O–W linkage for effective conversion of cellulose into ethylene glycol. Supplement to ACS Sustainable Chemistry & Engineering, 2020. URL <https://doi.org/10.1021/acssuschemeng.0c00836.s001>. Table S3.
- [33] Chris Torres. *Solvent and Surface Effects on Alkene Oxidation Catalysis over Transition Metal Incorporated Zeolites*. PhD thesis, University of Illinois Urbana-Champaign, 2023. URL <https://www.ideals.illinois.edu/items/128405>. Chapter 2, Figure 2.6.

- [34] Ohsung Kwon, E. Zeynep Ayla, David S. Potts, and David W. Flaherty. Influence of Ti-incorporated zeolite topology and pore condensation on vapor phase propylene epoxidation kinetics with gaseous H_2O_2 . *Angewandte Chemie International Edition*, 63:e202405950, 2024. doi: 10.1002/anie.202405950. URL <https://doi.org/10.1002/anie.202405950>.
- [35] Mikalai A. Artsiusheuski, Nicholas R. Jaegers, Carlos Lizandara-Pueyo, and Enrique Iglesia. Selective ethylbenzene dehydrogenation to styrene at lewis acid–base site pairs on zirconia surfaces. *ACS Catalysis*, 15:19102–19110, 2025. doi: 10.1021/acscatal.5c04904. URL <https://doi.org/10.1021/acscatal.5c04904>.
- [36] Tania L. Class-Martínez, Subramanian Prasad, Ahmad Moini, William F. Schneider, and Rajamani Gounder. Consequences of non-mean-field transport and agglomeration of extra-framework aluminum moieties on CHA zeolite dealumination rates. *Journal of Catalysis*, 458: 116848, 2026. doi: 10.1016/j.jcat.2026.116848. URL <https://doi.org/10.1016/j.jcat.2026.116848>.
- [37] Jinyong Luo, Krishna Kamasamudram, Neal Currier, and Aleksey Yezerets. NH_3 -TPD methodology for quantifying hydrothermal aging of Cu/SSZ-13 SCR catalysts. *Chemical Engineering Science*, 190:60–67, 2018. doi: 10.1016/j.ces.2018.06.015. URL <https://doi.org/10.1016/j.ces.2018.06.015>.
- [38] Seung-Hee Ryu, Seungeun Kim, Hyunjoo Lee, Joon-Hwan Choi, and Hojin Jeong. Robust palladium oxide nano-cluster catalysts using atomic ions and strong interactions for high-performance methane oxidation. *Nature Communications*, 15:8348, 2024. doi: 10.1038/s41467-024-52698-4. URL <https://doi.org/10.1038/s41467-024-52698-4>.
- [39] Paavo Auvinen, Niko M. Kinnunen, Janne T. Hirvi, Teuvo Maunula, Kauko Kallinen, Matthew Keenan, and Mika Suvanto. Effects of NO and NO_2 on fresh and SO_2 poisoned methane oxidation catalyst—harmful or beneficial? *Chemical Engineering Journal*, 417:128050, 2021. doi: 10.1016/j.cej.2020.128050. URL <https://doi.org/10.1016/j.cej.2020.128050>.
- [40] Kai Wu, Vita A. Kondratenko, Mingxia Zhou, Dmitry E. Doronkin, Stephan Bartling, Qiyang Zhang, Shanlei Han, Xin Jia, Qi Liu, Dong Xiong, Chunming Xu, Guiyuan Jiang, Dan Zhao, Uwe Rodemerck, David Linke, and Evgenii V. Kondratenko. The role of reducibility of PtGaO_x -based catalysts for efficient and durable propane dehydrogenation. *Angewandte Chemie International Edition*, 64:e202506704, 2025. doi: 10.1002/anie.202506704. URL <https://doi.org/10.1002/anie.202506704>.
- [41] Avram M. Buchbinder, John J. Senetar, Richard A. Johnson, II, Sathit Kulprathipanja, and Wei Pan. Process for regenerating a dehydrogenation catalyst. Patent application US20220126281A1, 2022. URL <https://patents.google.com/patent/US20220126281A1/en>. Description and Examples 1–7.
- [42] Sabah El Mohammad, Noémie Noël, Olivier Proux, Antonio Aguilar Tapia, Jean-Louis Hazemann, Christèle Legens, Céline Chizallet, and Kim Larmier. Identification of key active species in the conversion of C6 and C4 sugars by tungsten and molybdenum homogeneous salts. *ACS Catalysis*, 14:10998–11013, 2024. doi: 10.1021/acscatal.4c02791. URL <https://doi.org/10.1021/acscatal.4c02791>. Supporting Information, Figures S7–S8.